NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
(KARACHI CAMPUS)
FAST School of Computing
Fall 2026



AI4009 Generative AI: Course Project Proposal

When Does a GAN Help? Comparing GAN-Generated Retinal Images Against Simple Fixes for Rare Severe Diabetic Retinopathy Grades

TEAM MEMBERS:
Nabira Khan - 23K-0914

SECTION:
BCS-7A

INSTRUCTOR:
Dr. Usman Ali

PROBLEM STATEMENT

Diabetic retinopathy (DR) is graded from 0 (none) to 4 (proliferative). In public datasets the severe grades are rare. The APTOS 2019 training set has 3,662 images, of which only 193 are grade 3 and 295 are grade 4 (about 13% together). Classifiers trained on such data miss exactly the cases that need urgent referral. GANs are often proposed to generate extra images of the rare grades, but in the studies I reviewed they were compared mainly with no augmentation or weak augmentation, were tested only on the dataset they were trained on, and did not show how the benefit changes as the amount of real data shrinks or how many synthetic images should be added. It is therefore unclear when a GAN is worth its cost compared with cheap fixes such as class weights, oversampling and modern augmentation.

MOTIVATION AND OBJECTIVES

A clear answer helps anyone building medical classifiers with little data, and a negative result is just as useful as a positive one. The project has five objectives:

- Train a class-conditional StyleGAN2-ADA generator on all five DR grades of APTOS 2019, so that the rare severe grades (3 and 4) can borrow retinal structure from the common grades, and judge its output with FID, KID and visual inspection.
- Measure whether adding GAN images of grades 3 and 4 improves a classifier on those grades, compared with five simple baselines, including a strong RandAugment plus mixup baseline [9], [10].
- Map when the GAN helps along two axes: the amount of real rare-grade data (10%, 25%, 50% and 100%) and the number of synthetic images added (0.5, 1 and 2 times the real rare-grade count).
- Test whether any gain survives a change of clinic and camera by evaluating all classifiers, without retraining, on the IDRiD grading set [11].
- Check the synthetic images for responsible use: pigmentation diversity (using the Retinal Pigmentation Score [5]), memorisation of training images, and artefacts.

SELECTED MODEL AND REASON

The model is StyleGAN2-ADA, a GAN with adaptive discriminator augmentation. ADA was designed to stop the discriminator from overfitting when the training set holds only a few thousand images or fewer, which is the situation here [1]. It builds on StyleGAN2 [2] and supports class-conditional training, so one generator can produce any chosen grade. It is a GAN, so it fits the course rule, and a pretrained checkpoint can be fine-tuned, which suits a Kaggle T4.

REFERENCE PAPER

Full reference: [1] T. Karras, M. Aittala, J. Hellsten, S. Laine, J. Lehtinen, and T. Aila, "Training generative adversarial networks with limited data," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020, pp. 12104-12114. [Online]. Available: https://arxiv.org/abs/2006.06676

Summary: The paper shows that GAN training on small datasets fails because the discriminator memorises the training images. It applies a wide set of differentiable augmentations to every image the discriminator sees, and adjusts how often they are applied automatically, based on a measure of overfitting. This stabilises training and greatly improves FID with a few thousand images.

Part to implement: the ADA training procedure, applied to class-conditional retinal image generation. I will use NVIDIA's official PyTorch code (NVlabs/stylegan3, which includes the StyleGAN2-ADA configuration and runs on current PyTorch versions [7]) and acknowledge it clearly.

My own work is the data pipeline, the training runs on APTOS, the ablation of ADA on versus off at the smallest data size, the classifier experiments and baselines, the synthetic-ratio sweep, the external IDRiD test, the pigmentation and memorisation checks, the analysis and an interactive demo.

Planned deviations: images at 256 by 256 pixels instead of the paper's larger sizes, fine-tuning from a released pretrained StyleGAN2 checkpoint (with a newly initialised class-embedding layer) instead of training from scratch, and far fewer training iterations, all because of the T4 limit. Results will therefore not be compared one-to-one with the paper's FID values.

DATASET, TOOLS AND COMPUTE

- **Data:** APTOS 2019 Blindness Detection (Kaggle), 3,662 color fundus photos with DR grades 0 to 4, used for all training. Competition terms apply and will be checked before use [6]. IDRiD [11] (516 graded fundus photos from one clinic in India, CC BY 4.0) is used only as an external test set. No data is redistributed.
- **Tools:** Python, PyTorch, NVlabs stylegan3 code in its StyleGAN2-ADA configuration [7], torchvision (ResNet-18 or EfficientNet-B0 classifier, RandAugment), scikit-learn, the public Retinal Pigmentation Score code [8], and Gradio for the demo app.
- **Compute:** Kaggle notebooks with one T4 GPU, within the weekly GPU quota. Images are resized and cached once, GAN runs are short and checkpointed so they can resume across sessions, and classifiers use mixed precision.

IMPLEMENTATION AND EVALUATION PLAN

Data is split into train, validation and test sets with stratification by grade. APTOS has no patient identifiers, so I will remove near-duplicate images with perceptual hashing and keep the test set untouched by the GAN. In each experiment, grades 0 to 2 are kept in full and only the allowed fraction of grade 3 and 4 training images is kept. The conditional GAN is fine-tuned only on that reduced training split, so no test information reaches it. IDRiD is never used for training or tuning.

For each data fraction (10%, 25%, 50%, 100%), a classifier is trained under six conditions: (1) no augmentation, (2) class weights, (3) oversampling, (4) classical augmentation (flips, rotations, color jitter), (5) strong augmentation (RandAugment plus mixup), and (6) strong augmentation plus GAN images of grades 3 and 4, added at 1 times the real rare-grade count. At 10% and 100%, condition (6) is repeated with 0.5 and 2 times as many synthetic images. Each setting uses 3 random seeds, for 84 classifier runs in total.

- **Classifier metrics:** macro F1, quadratic weighted kappa (the APTOS metric), and recall for grades 3 and 4, on the APTOS test split and on IDRiD.
- **Generator metrics:** FID and KID against held-out real images, plus a visual check of diversity and artefacts. Because the real set is small, these numbers are reported as rough indicators only, and FID alone is known to be an unreliable guide for retinal images [3].
- **Ablation:** StyleGAN2 with and without ADA at the smallest data size, to test the paper's central claim on this data.
- **Baseline:** conditions (1) to (5) are the baselines. The main question is whether (6) beats (5), the strongest cheap fix, and how the answer changes with the real-data fraction, the synthetic ratio and the test dataset.

TIMELINE, EXPECTED OUTPUT AND RISKS

Week	Plan
1	Download APTOS and IDRiD, check terms, clean and deduplicate, split, resize and cache images.
2	Train and tune the classifier and the five simple baselines on full data, set up the StyleGAN2-ADA code.
3	Fine-tune the conditional StyleGAN2-ADA on all grades, first samples, FID and KID.
4	GAN runs for each data fraction, ADA ablation, memorisation check.
5	Learning-curve and synthetic-ratio experiments, 3 seeds per setting, IDRiD evaluation.
6	Run the Retinal Pigmentation Score on real and generated images, collect all results.
7	Analysis, limitations, responsible-AI section, Gradio demo app.
8	Final report, clean code and README, 10-minute demo rehearsal.

Expected output: working notebooks that train the generator and the classifiers, a set of generated DR images labelled synthetic, a results table, a learning-curve plot and a real-data versus synthetic-ratio map comparing all six conditions on APTOS and IDRiD, a Gradio demo app (generate an image of a chosen grade next to its nearest real training image, compare the baseline and GAN-augmented classifiers with Grad-CAM on an uploaded image, and browse the results), and a short written answer to when the GAN helps.

Key risks: (a) With only a few dozen rare-grade images at the 10% fraction, the GAN may produce poor images even with conditioning. Fallback: train at 128 by 128, or report the failure and the ADA ablation as the finding. (b) The GAN may not beat strong augmentation, which is still a valid result. (c) T4 time may run out, so training runs are short and checkpointed, and the synthetic-ratio sweep is limited to two fractions. (d) The pigmentation score may fail on some images or show a narrow range in APTOS, and the number of unscored images will be reported.

ETHICAL, PRIVACY, BIAS AND COPYRIGHT CONSIDERATIONS

- Generated images are experimental and not clinically validated. They will be labelled synthetic and not used for diagnosis.
- Synthetic images are not automatically anonymous. A nearest-neighbour check against training images will look for memorised copies.
- A generator can reproduce the biases of its data. I will compare the pigmentation distribution of real and generated images, since earlier work has shown that retinal AI can perform unevenly across pigmentation groups [4], and discuss that APTOS and IDRiD both come from India, so the external test checks a change of clinic and camera, not of population.
- APTOS is used under its Kaggle competition terms and IDRiD under CC BY 4.0. All datasets, code, libraries and pretrained weights will be cited, and reused components will be clearly marked.

REFERENCES

[1] T. Karras, M. Aittala, J. Hellsten, S. Laine, J. Lehtinen, and T. Aila, "Training generative adversarial networks with limited data," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020, pp. 12104-12114. [Online]. Available: https://arxiv.org/abs/2006.06676

[2] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of StyleGAN," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2020, pp. 8110-8119. [Online]. Available: https://arxiv.org/abs/1912.04958

[3] Y. Wu, F. Liu, R. Yilmaz, H. Konermann, P. Walter, and J. Stegmaier, "A pragmatic note on evaluating generative models with Fréchet Inception Distance for retinal image synthesis," in Proc. 9th Int. Conf. Med. Imaging Deep Learn. (MIDL), Proc. Mach. Learn. Res., vol. 315, 2026, pp. 2352-2368. [Online]. Available: https://proceedings.mlr.press/v315/wu26c.html

[4] P. Burlina, N. Joshi, W. Paul, K. D. Pacheco, and N. M. Bressler, "Addressing artificial intelligence bias in retinal diagnostics," Transl. Vis. Sci. Technol., vol. 10, no. 2, Art. no. 13, Feb. 2021, doi: 10.1167/tvst.10.2.13.

[5] A. E. Rajesh et al., "Ethnicity is not biology: Retinal pigment score to evaluate biological variability from ophthalmic imaging using machine learning," medRxiv, preprint, 2023, doi: 10.1101/2023.06.28.23291873.

[6] "APTOS 2019 blindness detection," Kaggle. Accessed: Oct. 19, 2026. [Online]. Available: https://www.kaggle.com/competitions/aptos2019-blindness-detection

[7] NVlabs, "stylegan3: Official PyTorch implementation of StyleGAN3, including the StyleGAN2-ADA configuration," GitHub. Accessed: Oct. 19, 2026. [Online]. Available: https://github.com/NVlabs/stylegan3

[8] uw-biomedical-ml, "retinal-pigmentation-score," GitHub. Accessed: Oct. 19, 2026. [Online]. Available: https://github.com/uw-biomedical-ml/retinal-pigmentation-score

[9] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "RandAugment: Practical automated data augmentation with a reduced search space," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2020, pp. 702-703. [Online]. Available: https://arxiv.org/abs/1909.13719

[10] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in Proc. Int. Conf. Learn. Represent. (ICLR), 2018. [Online]. Available: https://arxiv.org/abs/1710.09412

[11] P. Porwal et al., "Indian Diabetic Retinopathy Image Dataset (IDRiD): A database for diabetic retinopathy screening research," Data, vol. 3, no. 3, Art. no. 25, 2018, doi: 10.3390/data3030025.